OLUBUNMI E. OGUNLEYE

Email: oeogunle@cougarnet.uh.edu, LinkedIn, GitHub, Telephone: 832-716-6616

EDUCATION

University of Houston, Texas

PhD in Civil Engineering, **August 2025 – May 2029**

Advisor: Dr. Abigail L. Beck GPA: 3.67/4.0

Research Focus: Infrastructure and Community Resilience Assessment

Bauhaus-Universität Weimar, Germany

Master of Science in Digital Engineering **September 2023 – July 2025**

Advisor: Prof. Stefan Kollmannsberger GPA: 2.0/1.0

Thesis: A Hybrid Variational Autoencoder and Convolutional Neural Network Framework for Diagnosing Structural Damage and Sensor Faults.

Osun State University, Osogbo, Nigeria

Bachelor of Engineering in Civil Engineering **November 2014 – September 2019**

Advisor: Prof. Simon Olawale GPA: 4.85/5.0

Thesis: Parametric Investigation of the Reliability of One-way Reinforced Concrete Slab

RESEARCH INTEREST

- Probabilistic Fragility and Resilience Assessment of Infrastructure Systems.
- Risk-Informed Decision Support for Infrastructure Network Safety and Operation.
- Restoration and Recovery Modeling at the Infrastructure and Community Level.
- Infrastructure Asset Management and Lifecycle Planning.
- Digital Twins and Data-Driven Condition Monitoring of Infrastructure Systems.

RESEARCH EXPERIENCE

Beck Research Group – University of Houston

Graduate Research Assistant **August 2025 - Present**

Duties:

- Probabilistic Assessment of Electrical Power Distribution Network (EPDN) in Houston and Galveston Subjected to wind hazards.
- Improving methods to develop fragility models for characterizing the vulnerability of EPDN at the component and network levels.
- Integrate asset inventory and hazard data to support risk-informed infrastructure management decisions.
- Improving models to estimate loss, resilience, and recovery of EPDN under natural hazards.

Chair of Data Engineering in Construction – Bauhaus Universität Weimar, Germany

Research Assistant **March 2025 – August 2025**

Duties:

- Developed and validated a Long Short-Term Memory Variational Autoencoder for unsupervised anomaly detection and a Convolutional Neural Network for fault classification in structural health monitoring.
- Simulated multi-DOF structural systems with sensor faults and stiffness degradations to generate training and testing datasets.
- Validated framework on experimental sensor datasets from Open Lab Bridge in Dresden.

Chair of Mechanics of Materials – Bauhaus Universität Weimar, Germany

Research Assistant **March 2024 – August 2025**

Duties:

- Carried out research activities focused on rheological studies of cement paste.
- Literature research and formulation on smart and functional building materials.
- Additive Manufacturing using 3D printers to evaluate the performance of smart materials for construction.

Structural Engineering Laboratory – Osun State University

Undergraduate Research Assistant **November 2018 – February 2019**

Duties:

- Developed Advanced First Order Second Moment (AFOSM) framework for reliability assessment of infrastructures.
- Performed physical and Non-Destructive Testing on selected buildings and bridges within the city of Osogbo to gather data for reliability assessment.

WORK EXPERIENCE

Telecom Structural Engineer **August 2021 – September 2023**
KA Engineering Group, Belfast, UK
Duties:

- Analyzed simple and complex telecom transmission structures, including lattice towers, wall-mounted, and roof-mounted telecom structures, utilizing fundamental engineering principles and finite element analysis techniques.
- Assessed the structural stability of these structures and prepared calculation reports, designs, and drawings.
- Provided technical advice and recommendations to clients regarding safe designs and constructions.

Structural Engineer **May 2021 – July 2021**
Efficacy Construction Company, Lagos, Nigeria
Duties:

- Conducted structural analysis and design for residential blocks and units in preparation for construction.
- Provided site supervision and monitoring to ensure the implementation of design specifications and recommendations.

Structural Engineer **April 2020 – April 2021**
Dillon Consultants Nigeria Limited, Lagos, Nigeria
Duties:

- Conducted structural analysis and design of multistory reinforced concrete and steel structures and their foundation in compliance with codes and client specifications.
- Produced reports, calculation sheets, and detailed drawings for construction purposes.
- Supervised construction activities on-site to ensure adherence to design specifications.

PUBLICATIONS

- Olawale S.O.A, **Ogunleye O.E.,** Adekilekun T.M. (2019). Parametric Investigation of the Reliability of One-way Reinforced Concrete Slab. *In Conference Proceedings of the International Journal of Engineering and Environmental Sciences, Osun State University Edition 1, 2019* (Link).
- **Ogunleye O.E**., Adebanjo A.U., Shafiq N., Jegede A.V., Musa Umar Kolo (2026). Hybrid LSTM–VAE–CNN Framework for Structural Damage Prognosis in Prestressed Concrete Bridges [**Manuscript submitted to Springer Journal of Civil Structural Health Monitoring**].
- **Ogunleye O.E**., Adebanjo A.U., Alagbada V.A., Jegede A.V. (2026). Physics-Guided Deep Operator Network for Spatial Displacement Reconstruction of a Prestressed Concrete Bridge from Sparse Load-Test Measurements. *Structures*. **[Manuscript under revision].**
- Adebanjo A.U., Aliyu Y., Shafiq N., **Ogunleye O.E.**, Kolo M.U., Adejumo D.F., Opadotun K., Muthusamy K. (2026). Explainable Modeling of Constituent Effects on the Compressive Strength of Low-Carbon Recycled Concrete. *Innovative Infrastructure Solutions* [**Manuscript under consideration**].
- Dayyabu A., **Ogunleye O.E.**, Kolo M.U., Amuda A.G., Jegede A.V., Adebanjo A.U. (2026). Explainable Physics-Informed Machine Learning for Predicting the Ultimate Bearing Capacity of Bored Piles in Layered Cohesive-Frictional Heterogeneous Soils. *Transportation Infrastructure Geotechnology* [**Manuscript under consideration; editor assigned**].

SKILLS AND COMPETENCIES

- **Computational and Analytical Skills:** ANSYS, AnyLogic, QGIS, StaadPro, ProtaStructures, AutoCAD 2D&3D, Revit, Robot, BIM 360, Navisworks, DesiteMD.
- **Digital and Programming Skills:** C#, Java, Python, MATLAB.
- **Machine / Deep Learning**: Foundation Models, Algorithm Development, Optimization, Probabilistic Models.
- **Software Development**: Requirement Engineering, Software Development and Testing, Agile Methods, Git/Github, Gradle, SQLite/SQLAlchemy.

FELLOWSHIPS, GRANTS AND AWARDS

- German Government DAAD Masters Scholarship. **2024**
- Nigerian Government Petroleum Technology Development Fund Postgraduate Scholarship - Research on Digital Monitoring of Energy Infrastructure. **2023**
- Nigerian National Petroleum Postgraduate Scholarship **2022**
- First-class graduate award and overall best graduating student at Osun State University. **2019**
- Winner, "Bridge to Better Life" Design Competition by Nigerian Institute of Structural Engineers. **2019**

ACADEMIC AFFILIATION, LEADERSHIP, AND COMMUNITY SERVICE

- Graduate Student Member, American Society of Civil Engineers **2026**
- Member, Graduate Student Council, NSF Natural Hazard Engineering Research Infrastructure (NHERI) **2025**
- Member, Blacks in AI. **2022**
- Graduate Member, Nigerian Society of Engineers. **2020**
- Graduate Member, Institute of Structural Engineers. **2020**
- Academic Coordinator, Nigerian Institute of Civil Engineers – Student Affiliate **2019**
  Osun State University, Osogbo, Nigeria.
- Volunteered and designed a low-cost bridge linking two communities in Osogbo, Osun State Nigeria **2019**

REFERENCES

Available on Request.